

Game Makers — Week 7

Operators, My Blocks and Levels

Name: _____

TODAY'S BIG IDEA

Two ideas today.

Operators let you do maths and ask complicated questions. `(score) + 10` · `(score) > 10` · `<this>` and `<that>` · `join [Level ] (level)`

My Blocks let you take a group of blocks you keep rewriting, give it a name, and use it as one single block from then on.

"How do you make a cup of tea?" — seven steps. Draw a box around them and call it **MAKE TEA**. Now "make tea for four people" is `repeat 4 [MAKE TEA]` instead of 28 steps.

That's called **abstraction**, and it's one of the most powerful ideas in all of programming.

WORDS TO KNOW

Word	What it means
operator	A block that does maths or asks a question
My Block	A block you invent yourself, made of other blocks
define	The hat block where you say what your new block does
abstraction	Naming a group of steps so you can forget the details

💡 **Dragging tip:** building `<(score) > 10>` and `<(lives) > 1>` inside a tiny slot is horrible. Build the whole thing in an **empty space** first, then drag the finished assembly into the `if`.

CLASS ACTIVITY A — QUIZ SPRITE

```
ask [What is 7 x 6?] and wait
if <(answer) = 42> then
  say [Correct!] for 2 seconds
  change score by 1
else
  say (join [No, it was ] [42]) for 2 seconds
```

Add two more questions. Notice how much you're copying and pasting. Activity B fixes that.

CLASS ACTIVITY B — MAKE YOUR OWN BLOCK

```
define celebrate
  say [Correct!] for 1 second
  change size by 25
  start sound [Cheer]
  wait 0.5 seconds
  change size by -25
```

Now replace all three copies of your celebration with one `celebrate` block. **Change the definition once → all three change.** That's the payoff.

YOUR PROJECT — IT GETS HARDER AS YOU GET BETTER

- [] A `speed` variable. Replace the fixed `wait 0.02` with `wait (speed) seconds`
- [] Every 10 points: level up and multiply `speed` by 0.85

```
forever
  if <(score) > ((level) * 10)> then
    change level by 1
    set speed to ((speed) * 0.85)
    say (join [Level ] (level)) for 1 seconds
  wait 0.1 seconds
```

- [] A **rare bonus item** worth 5 points that only shows up sometimes:

```
if <(pick random 1 to 5) = 1> then
  show
else
  hide
```

- [] Make a `reset object` My Block and use it on every falling sprite
 - [] **Save it**
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Homework

1. **A rare treasure.** If you didn't finish the bonus item, do it now.
 2. **Tidy up.** Find code you've written more than once and turn it into a My Block.
 3. **Think about the showcase.** Next week your family comes to see your game. Have two sentences ready: *what your game is*, and *the hardest bug you fixed*.
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Bonus challenges

1. A power-up that makes your player wider for five seconds, then back to normal.

2. Use `join` to show "Level 3 — Score 42" as one readout.
 3. Use `and` or `or` for a combo condition — a bonus that only counts on a high level.
 4. **Really hard:** replace your duplicated falling sprites with **clones**. Look up `create clone of myself`, `when I start as a clone`, `delete this clone`. Nobody is expected to get this one.
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IF SOMETHING BREAKS

- **Game freezes on level up?** `speed` hit zero or went negative. Put a floor on it.
 - **Level goes up hundreds of times?** The `if` keeps firing. Make sure `level` is in the condition.
 - **Can't drop an operator in?** Build it in empty space first, then drag it in.
 - **My Block missing on another sprite?** My Blocks belong to one sprite only.
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